PLATFORMS AND TOOLKITS FOR IoT AND EDGE COMPUTING

Software	URL	Usage
Node-RED	nodered.org	Flow based programming environment
Contiki	contiki-os.org	Microcontrollers, IPv6, IPv4, protothreads, low resource, game consoles
FlowHub	flowhub.io	Flow based IoT programming
NoFloJS	noflojs.org	JavaScript based flow programming
Netron	github.com/lutzroeder/netron	Dynamic visualisation
PyFlow	wonderworks-software.github.io/PyFlow	Visual scripting
Yet another robot platform (YARP)	yarp.it	Robotic programming
OROCOS	orocos.org	Robotic programming and machine control
OpenIoT	openiot.eu	Sensing as a service (S2aaS)
Zetta	zettajs.org	WebSocket programming, real-time on TCP, reactive programming, low overhead scenarios
DSA	iot-dsa.org	Real-time interfacing, inter-device communication, programming on multiple layers
IoTivity	iotivity.org	Constrained application protocol (CoAP), IoT programming
CupCarbon	cupcarbon.com	Smart city, SCI-WSN simulation, visualisation, 2D and 3D OpenStreetMap, MQTT programming, sensor programming
KAA	kaaproject.org	Data analytics, real-time applications, dynamic communications and updates

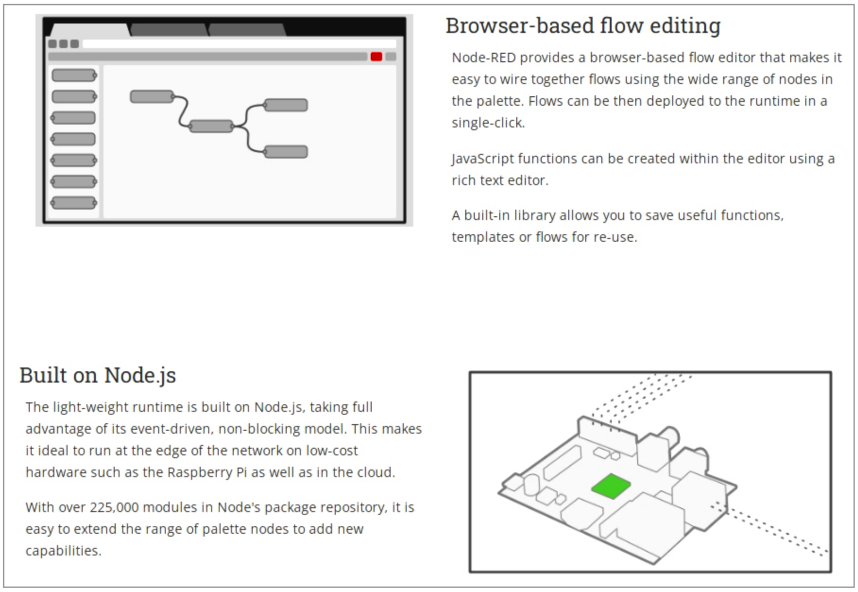

Fig. 2: Node-RED platform built on Node.js (Source: nodered.org)

gadgets, theft avoidance, controlling home appliances, etc.

Retail sector. Automated checkout, logistics monitoring and management, etc.

Agriculture. Analysis of crops, dynamic water distribution, smart irrigation, farm surveillance, drones for smart farming, agriculture robots, etc.

Many other domains, too, are associated with the field of IoT, especially where intelligent robotic applications are being developed. The Internet of Everything (IoE) is another term used for smart applications, and is the integration of IoT with the cloud and the World Wide Web for the real-time connectivity of devices.

A number of programming platforms are available for working with IoT, IoE, fog or edge scenarios. Hundreds of toolkits, which are very powerful and easy to use for dynamic research, are available as well. A few of these are listed in the Table.

Node-RED: A tool for flow based programming of IoT scenarios

Node-RED (*https://nodered.org*) is a powerful and easy-to-use programming platform for the simulation of IoT scenarios. Fog and edge computing can also be done using flow based programming in Node-RED. Here, high performance structures can be implemented using minimum coding.

Installation and working

Node-RED is a specialised package that is installed onto the Node.js platform. The latter is a lightweight but high performance programming environment based on JavaScript. Many packages are available in Node.js for multiple applications, including Internet of Things (IoT), cloud computing, machine learning, data science, and blockchain.

To work with Node-RED, the Node.js platform should be installed first, which is available at *https://nodejs.org* for multiple operating systems like Windows, Mac, and Linux for 32-bit or 64-bit architectures (Fig. 1).

After the installation of Node.js, the Node-RED package can be installed from the node package manager (NPM), which is the repository of packages developed and deployed